PROVEIT RESEARCH REPORT

Selective coefficient extraction

for explicit ZFC consistency equations

A smaller polynomial in ten unrestricted integers,
and a companion polynomial in nine positive integers

Witness domain	Exact degree	Binary operations
$\mathbb{Z}^{10}$	35 876 184	640
$\mathbb{Z}^{9}_{>0}$	19 615 180	606
$\mathbb{N}^{9}$, with $0 \in \mathbb{N}$	19 615 180	609

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Repository snapshot:
`124c07c0b77a9e20e84676514d8b1ff78536ea3f`.

Evidence level. Explicit coefficient data, complete finite arithmetic circuits, ordinary mathematical proofs of the new reductions, and independent finite audits. The unbounded compiler and number-theoretic theorems are named dependencies. This report does not claim a new end-to-end proof-assistant certificate or a global simplicity minimum.

Abstract

We construct an explicit integer-coefficient polynomial in ten unrestricted integer variables whose lack of solutions is equivalent, by an arithmetic argument formalizable in ZFC, to syntactic consistency of ZFC. Its exact total degree is 35 876 184 and a literal division-free evaluator has 640 binary operations. Relative to the current package-40 ten-integer construction at the reviewed repository snapshot, the degree is smaller by a factor of 6.62 and the evaluator has 142 fewer operations. A companion Jones construction uses nine positive variables, degree 19 615 180, and 606 operations.

The improvement comes from changing the coefficient-extraction problem. The source contains 22 quadratic equations in 31 positive variables, but only 62 of the 528 possible padded quadratic monomials occur. Their positions must be isolated; positions of unused monomials may collide. An explicit weight list of span 175 suffices, instead of the inherited Sidon span 799. Positivity is applied once to the common packed word, avoiding expansion of every source equation. Nonuniform target positions and removal of common padding further shorten the exponent range. Exact symbolic audits verify every extracted quadratic equation. All coefficients are supplied by finite recipes and authenticated binary payloads; none is chosen by an unspecified existence principle. The work preserves the inherited compiler tuple and attributes the nine- and ten-variable arithmetic bridges to their sources.

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1 Results and their precise meaning

Let Con(ZFC) mean that there is no finite derivation of falsity from the repository's fixed, effectively presented ZFC axiom predicate. We use $\mathbb{N} = \{0, 1, 2, \ldots\}$ and distinguish positive, nonnegative, and unrestricted integer witnesses throughout.

Theorem 1.1 (Explicit consistency equations)**.** *The coefficient data and substitution rules in this report define fixed polynomials $P_{10} \in \mathbb{Z}[f, g, h, k, l, w, x, y, z, m]$ and $P_9 \in \mathbb{Z}[X_1, \ldots, X_9]$. Using the pinned positive-parent compiler theorem and the arithmetic bridges specified in Sections 2, 6, and 8, the following equivalences have proofs formalizable in ZFC:*

$$\neg\exists \mathbf{x} \in \mathbb{Z}^{10}\ P_{10}(\mathbf{x}) = 0 \quad \longleftrightarrow \quad \mathrm{Con(ZFC)}, \tag{1}$$

$$\neg\exists \mathbf{x} \in \mathbb{Z}^{9}_{>0}\ P_{9}(\mathbf{x}) = 0 \quad \longleftrightarrow \quad \mathrm{Con(ZFC)}. \tag{2}$$

The exact degrees are respectively 35 876 184 *and* 19 615 180*. The supplied evaluators use respectively 640 and 606 binary additions, subtractions, and multiplications. The nine-nonnegative version has the same degree and 609 operations.*

These are unconditional equivalences: their proofs do not assume that ZFC is consistent or inconsistent. They do not decide either side. The precise polynomial is specified by its finite arithmetic circuit and fixed leaves; an expanded list of millions of monomials is unnecessary. All displayed auxiliary names denote polynomial substitutions, not additional witnesses.

The comparison is to the current repository snapshot, including newly placed wave-6 packages 38–45. The supplied incoming directory contains its placement ledger; those archives have already been placed in the report collection. The active research beside the collection also contains later checked local degree reductions. These were included in the review [1, 2, 5, 6].

Construction	Domain	Exact degree	Operations
Package 40	$\mathbb{Z}^{10}$	237 520 456	782
This report, P_{10}	$\mathbb{Z}^{10}$	35 876 184	640
Package 45	$\mathbb{Z}^{9}_{>0}$	129 875 980	788
This report, P_9	$\mathbb{Z}^{9}_{>0}$	19 615 180	606
Inherited P67	$\mathbb{Z}^{15}_{>0}$	155	67
Checked local P75	$\mathbb{Z}^{15}_{>0}$	68	75
Local P74, Appendix A	$\mathbb{Z}^{15}_{>0}$	68	74

The main ten-integer degree decreases by 84.90%, while operation count decreases by 18.16%. The nine-positive degree decreases by 84.90%, and operations by 23.10%. P67 remains much shorter if operation count alone is prioritized and fifteen positive variables are acceptable. Thus "simplest" has no scalar answer without specifying a domain and an ordering of variable count, degree, arithmetic size, and coefficient size. No theorem of global optimality, and no new universal variable-count theorem, is claimed.

1.1 What is new

Three changes compose without altering the inherited machine or compiler:

1. Isolate only source-supported monomials. This replaces a full Sidon condition by a strictly weaker finite condition on 62 protected pairs.
2. Encode positive witnesses by adding a fixed word to the single common convolution factor. This avoids 31 separate successor substitutions in all 22 equations.
3. Choose target positions using the actual supports, then translate all targets by the maximal common amount that keeps exponents nonnegative.

The first two changes reduce both the necessary separation problem and the coefficient description. The third reduces degree without introducing new unknowns. Shared Horner evaluation turns these structural savings into the literal circuit counts above.

2 The fixed positive quadratic parent

The parent is the pinned 31-positive-variable, 22-quadratic-equation system retained in package 31 and independently recorded by package 40 [4, 3, 5]. Its variable order is

$$J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta.$$

The four fixed integer ports are β, K_0, M_C, M_F, and

$$q = \beta J + 1, \qquad k = \eta + \zeta, \qquad H = 4a + 3, \qquad C = q - F - \alpha$$

are affine abbreviations. All 31 witnesses here are strictly positive. The equations, in their original order, are

$$\begin{gathered}
L = q^2 - 1, \qquad N = q(L+1), \qquad X = wq, \\
a = E + sN, \qquad E = X(a - E), \qquad c = k(a - E) + \eta, \\
P = E(c - \eta) + k, \qquad \tau^2 = P(P - k) + 1, \\
D = X + ac + \gamma H, \qquad \Delta = a^2 + H, \qquad C_2 = c^2, \\
D^2 = \Delta C_2 + 1, \qquad k = jc - V + 1 + hE, \\
jc - V = (q - F)(N - q) - CL + (M_C + qM_F)J, \\
(K_0 + w)C + (q - F) - z(q - 1) = 1, \\
F_2 = f^2, \qquad V = of - c, \qquad g = y^2 - V^2, \\
I = iC_2, \qquad B_{\mathrm{aux}} = I^2, \qquad B_{\mathrm{aux}} = \Delta(F_2 - 1), \\
y^2 - B_{\mathrm{aux}} g = 1.
\end{gathered} \tag{3}$$

For numbering, read each line from left to right. Each residual r_j is left side minus right side, with the affine abbreviations expanded. Their complete sparse forms are supplied in `upstream/parent_sparse.json`. The finite auditor reconstructs the new weighted records from this data, keeping the four ports as independent symbolic indeterminates.

2.1 The inherited semantic theorem

The inherited ordinary mathematical theorem says that this positive system has a solution exactly when K215 halts on blank input, and hence exactly when the fixed ZFC proof predicate has a refutation. Its ingredients are the native-window/Pell compiler, the coherent coefficient calibration, the algebraic parent reduction, and the checked K215 endpoint [4, 3, 7, 8].

The present report does not change any of those ingredients. In particular, the positive transport coordinate z in (3) is the old, unshifted parent coordinate. The complete four-port tuple is inherited from package 31, with the mask convention already shifted by $B-1$. Mixing that tuple with components from another package would not be justified.

The active Lean research proves a nine-natural-variable existence theorem using a classically selected polynomial, and proves local equivalences of several explicit short circuits. It explicitly leaves the explicit native compiler bridge and an object-language ZFC derivation open [2]. Those are valuable separate results; they are not a new kernel certificate for the explicit polynomials constructed here.

2.2 Concrete coefficient data

The four native ports have the following exact bit lengths:

Port	Bits	Compressed payload
β	1 064 921	`coefficients/payloads/beta.bin.gz`
K_0	1 070 004	`coefficients/payloads/K.bin.gz`
M_C	1 064 921	`coefficients/payloads/MC.bin.gz`
M_F	1 064 922	`coefficients/payloads/MF_source.bin.gz`

The decompressed magnitude bytes are unsigned little-endian integers. Their total size is 533,099 bytes; the compressed total is 10,485 bytes. The native base is $2^{1\,064\,921}$ and β is this base minus one. The complete binding, hashes, reconstruction program, transition table, tile table, and inherited notices are included.

Only nine nonliteral coefficient recipes are needed for the new reversed kernel, in addition to a single source-base constant:

$$
\begin{aligned}
&c_0 = 2\beta, && c_1 = 2\beta^2, && c_2 = \beta,\\
&c_3 = M_F + M_C - 2\beta, && c_4 = 2\beta M_F - 2\beta^2, &&\\
&c_5 = 2K_0, && c_6 = K_0, && c_7 = K_0 + 1,\\
&c_8 = \beta(K_0 + 1). &&&&
\end{aligned}
$$

Their signs and occurrences are specified by the 94 records in `data/raw_recipes.json`. These are recipes for fixed integers, not extra variables. The original parent equations and the extraction rule also determine them uniquely, so the mathematical definition does not rely on trusting an opaque data file.

3 Selective extraction of quadratic coefficients

For a multi-index $\alpha \in \mathbb{N}^{31}$ with $|\alpha| \le 2$, pad its index multiset with $2 - |\alpha|$ copies of a dummy index 0. There are $32 \cdot 33/2 = 528$ possible padded pairs. Let $\mathcal{S}$ be the union of the monomial supports of the unshifted residuals. It consists of one constant, 22 linear monomials, and 39 quadratic monomials: 62 in all.

Definition 3.1. A list $w_0 = 0, w_1, \ldots, w_{31}$ is *selective for* $\mathcal{S}$ if the padded weight sum of each member of $\mathcal{S}$ is attained by exactly its own pair among all 528 possible pairs. Pair sums belonging exclusively to monomials outside $\mathcal{S}$ may coincide.

This differs materially from merely separating the 62 protected pairs from each other: each must also avoid all unused pairs. The latter check prevents an unused monomial from contaminating an extracted coefficient.

The supplied list is

$$\begin{aligned}(w_0, \ldots, w_{31}) = (&0, 175, 169, 108, 60, 123, 81, 158,\\ &139, 78, 63, 166, 164, 37, 12, 130,\\ &40, 174, 50, 21, 56, 14, 59, 76,\\ &101, 150, 143, 15, 149, 57, 146, 147).\end{aligned} \tag{4}$$

The ordering is the parent variable ordering, not increasing order. Its maximum is $t = 175$. It has only 278 distinct pair sums, so it is deliberately not a Sidon set. Nevertheless every one of the 62 protected pairs is isolated. The exhaustive finite certificate is included and checked independently from the search that found the list.

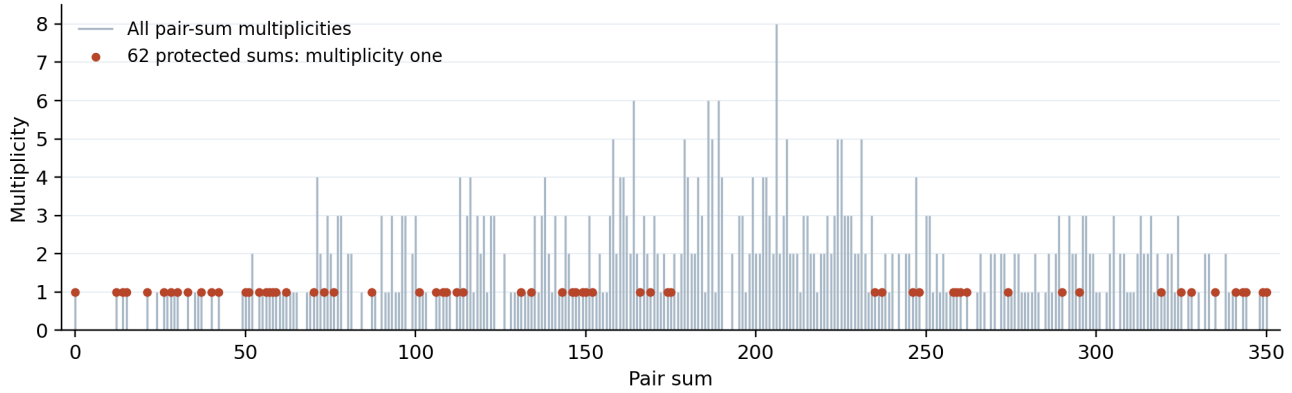


Figure 1: Pair-sum multiplicities of the explicit weights. Protected sums have multiplicity one; unused sums may have larger multiplicity. The certificate checks the full finite family, not just the displayed plot.

3.1 Targets selected for actual supports

For each original row, use the target and sign in Table 1. Only row 8 is negated. Write ρ_j for the resulting signed residual in increasing target order, and E_j for its target.

Write $e(\alpha) = \sum_i \alpha_i w_i$ and let $\mathcal{Q} = \{w_a + w_b : 0 \le a \le b \le 31\}$. The certificate checks, for every supported monomial of row i and every distinct target row j,

$$E_j - E_i \notin \mathcal{Q} - e(\alpha). \tag{5}$$

It also checks $E_i - e(\alpha) \ge 0$. All 1,974 cross-row exclusions pass. No uniform gap of $2t + 1$ is required.

With $\rho_j(\mathbf{p}) = \sum_\alpha a_{j,\alpha} \mathbf{p}^\alpha$, define

$$\mathcal{D}(Z) = \sum_{j,\alpha} \underbrace{(2 - |\alpha|)! \prod_i \alpha_i!\, a_{j,\alpha}}_{c_{j,\alpha}} Z^{E_j - e(\alpha)}. \tag{6}$$

Original row	Sign	Target	Original row	Sign	Target
16	+	246	13	+	2721
17	+	429	14	+	3063
18	+	657	12	+	3312
21	+	868	4	+	3506
6	+	1058	22	+	3798
2	+	1358	11	+	3989
19	+	1550	20	+	4179
7	+	1734	5	+	4367
3	+	2058	15	+	4623
9	+	2248	10	+	4817
1	+	2545	8	−	5148

Table 1: Exact target assignment for the 22 original quadratic residuals.

The sum has exactly 94 distinct powers of Z. Its least exponent is 0, its greatest is $E_* = 5148$, and its leading coefficient is 2. The last fact follows because the last row is $P(P-k)+1-\tau^2$, whose constant is 1.

Lemma 3.2 (Exact simultaneous extraction)**.** *For independent indeterminates $p_1, \dots, p_{31}$, put*

$$\mathcal{C}_{\mathbf{p}}(Z) = \left(1 + \sum_{i=1}^{31} p_i Z^{w_i}\right)^2.$$

Then for every j,

$$[Z^{E_j}]\,\mathcal{C}_{\mathbf{p}}(Z)\mathcal{D}(Z) = 2\rho_j(\mathbf{p}). \tag{7}$$

Proof. At a protected exponent $e(\alpha)$, selective isolation makes the coefficient of $\mathcal{C}_{\mathbf{p}}$ exactly $2!\mathbf{p}^{\alpha}/((2-|\alpha|)!\prod_i \alpha_i!)$. This cancels the factorial multiplier in (6). Condition (5) excludes every contribution from another row. These two statements account for every term of the convolution. □

Different records in (6) cannot have the same exponent. Within a row this follows from selective isolation. Across rows an overlap $E_i - e(\alpha) = E_j - e(\beta)$ would give the forbidden difference $E_j - E_i = e(\beta) - e(\alpha)$, with $e(\beta) \in \mathcal{Q}$. Consequently coefficient bounds do not acquire a factor of 22.

3.2 A shift of one word replaces 31 expanded substitutions

If a nonnegative digit word is $g = \sum n_i \mathcal{B}^{w_i}$, define $\ell = \sum \mathcal{B}^{w_i}$. Then

$$1 + \ell + g = 1 + \sum_i (n_i + 1)\mathcal{B}^{w_i}. \tag{8}$$

Thus one addition of ℓ evaluates the original parent at positive coordinates $p_i = n_i + 1$. Expanding $r_j(\mathbf{n}+\mathbf{1})$ before packing is unnecessary. That earlier expansion created 71 protected monomials, 136 coefficient records, and 17 nontrivial recipes. Retaining the raw parent leaves 62, 94, and 9 respectively.

3.3 Removing common padding

An initial valid target arrangement had first target 350 and last target 5252, with every kernel exponent at least 104. Subtracting 104 from all targets preserves all differences in (5). It gives the displayed

targets 246 through 5148 and makes the least kernel exponent zero. This is the maximal common translation for that arrangement.

The old package-40 product span was 36 775. The new one is

$$M = E_* + 2t = 5498. \tag{9}$$

The search is a discovery tool; only the finite list and its checked properties are premises of the proof. No optimality of span 175 or of the target arrangement is required.

4 A complete source for the ten-integer construction

Use the conservative fixed bound

$$H_0 = 8(1 + \beta + K_0 + M_C + M_F)^2, \qquad \beta_S = 2^{2\,140\,030}.$$

The finite audit finds ordinary coefficient ℓ^1-norm at most 4 and port degree at most two. Hence $|a_{j,\alpha}| \le H_0$ and each kernel coefficient has absolute value at most $2H_0$. Exact integer arithmetic verifies

$$\beta_S > 4 \cdot 33^2 H_0. \tag{10}$$

The bit lengths of H_0 and $4 \cdot 33^2 H_0$ are 2 140 010 and 2 140 022, respectively. The even exponent of β_S also ensures $\beta_S \equiv 1 \pmod 3$.

The letters below refer to the ten final unknowns, independently of similarly named parent coordinates. With integer f, g, define

$$\begin{aligned}
&b = 1 + 3f, && \mathcal{B} = \beta_S b^2, && \ell = \sum_i \mathcal{B}^{w_i},\\
&N_0 = \mathcal{B}^{176}, && N_1 = 4\mathcal{B}^{5499}, && N = N_0 N_1,\\
&M_c = N_0 - 1 - (b-1)\ell, && Q_c = (\mathcal{B}-2)\sum_j \mathcal{B}^{E_j}, &&\\
&T_c = M_c + Q_c N_0, &&&&\\
&J_c = 2(1+\ell+g)^2 \mathcal{D}(\mathcal{B}) + \mathcal{B}\sum_{r=0}^{5498} \mathcal{B}^r, &&&&\\
&S_c = g + J_c N_0, && R_c = (S_c + T_c + 1)N + T_c + 1, &&\\
&X = (N-1)R_c, && Y = N^2, && \mathcal{C} = b\mathcal{B}^{175}.
\end{aligned} \tag{11}$$

All sums and powers have fixed limits. In particular,

$$N = 4\mathcal{B}^{5675}, \qquad \mathcal{C} = 2^{374\,505\,250} b^{351}.$$

The latter has exactly the even power-of-two coefficient and odd b-exponent required by the inherited ten-integer bridge.

Proposition 4.1 (Source interface)**.** *The following statements hold.*

(i) If the positive parent has a solution, then for every sufficiently large positive f with b a square and a power of two there is an integer g such that $b \le g < \mathcal{C}$ and $Y \mid \binom{2X}{X}$.

(ii) If $f \ne 0$, b is a square and a power of two, $0 \le g < 2\mathcal{C}$, and $Y \mid \binom{2X}{X}$, the positive parent has a solution.

(iii) Already if $f \ne 0$, b is a square, and $0 \le g < 2\mathcal{C}$, without assuming it is a power of two, one has $0 \le S_c, T_c < N$, $3 \mid X$, $X > 3b$, and $Y > \max(b, 256)$.

4.1 Bounds before digit recovery

If $b = 1 + 3f$ is a square and $f \neq 0$, then $f > 0$, $b \geq 4$, and $\mathcal{B} > 2b$. Since the kernel has leading coefficient 2 and all other coefficients bounded by $2H_0$,

$$\mathcal{D}(\mathcal{B}) \geq 2\mathcal{B}^{E_*} - 2H_0 \sum_{r<E_*} \mathcal{B}^r > 0.$$

Its absolute value is at most $2H_0 \sum_{r=0}^{E_*} \mathcal{B}^r$. Distinct positive weights give $1 + \ell < 2\mathcal{B}^t$; hence for the full permitted interval $0 \leq g < 2b\mathcal{B}^t$,

$$1 + \ell + g < 3b\mathcal{B}^t.$$

It follows from (10) that

$$\begin{aligned} 0 < J_c &\leq (36H_0b^2 + \mathcal{B}) \sum_{r=0}^{M} \mathcal{B}^r \\ &< 2\mathcal{B}^{M+1} < N_1, \end{aligned}$$

where $36H_0b^2 \leq \mathcal{B} - 2$ suffices. Also $g < N_0$, so $0 < S_c < N$. The mask digits give $0 \leq M_c < N_0$; likewise $0 \leq Q_c < \mathcal{B}^{E_*+1} < N_1$, yielding $0 \leq T_c < N$.

Since $b \equiv \beta_S \equiv 1 \pmod 3$, one has $N \equiv 1 \pmod 3$, and thus $3 \mid X$. Moreover $R_c \geq N + 1 > \mathcal{B} > b$ and $N \geq 4$ imply $X > 3b$. The chosen base gives $Y > \max(b, 256)$. This proves (iii). The ordering of this argument matters: the arithmetic bridge recovers that b is a power of two only after these range estimates are available.

4.2 Two exact carry tests

Now assume b is a power of two. Then $\mathcal{B}, N_0, N$ are powers of two. For nonnegative integers let

$$\tau_2(u, v) = \mathrm{pc}(u) + \mathrm{pc}(v) - \mathrm{pc}(u + v)$$

be the number of binary carries, counted with multiplicity. Its vanishing means that the binary supports of u and v are disjoint.

The low mask M_c has digit $\mathcal{B} - b$ at the 31 weights and $\mathcal{B} - 1$ elsewhere below N_0. Therefore

$$\tau_2(g, M_c) = 0 \quad \Longleftrightarrow \quad g = \sum_i n_i \mathcal{B}^{w_i}, \quad 0 \leq n_i < b. \tag{12}$$

Set $p_i = n_i + 1$. The coefficients of $\mathcal{C}_{\mathbf{p}}\mathcal{D}$ have absolute value at most $2H_0(1 + \sum p_i)^2 < \mathcal{B}/2$ by $p_i \leq b$ and (10). Consequently

$$K_c = (1 + \ell + g)^2 \mathcal{D}(\mathcal{B}) + \frac{\mathcal{B}}{2} \sum_{r=0}^{M} \mathcal{B}^r$$

has ordinary base-$\mathcal{B}$ digits strictly between 0 and $\mathcal{B}$. The digit at target E_j is $\mathcal{B}/2 + 2\rho_j(\mathbf{p})$. For such a digit d, disjointness from $\mathcal{B}/2 - 1$ is equivalent to $d = \mathcal{B}/2$. Since $J_c = 2K_c$ and $Q_c = 2(\mathcal{B}/2 - 1) \sum_j \mathcal{B}^{E_j}$, a common binary shift gives

$$\tau_2(J_c, Q_c) = 0 \quad \Longleftrightarrow \quad \rho_1(\mathbf{p}) = \cdots = \rho_{22}(\mathbf{p}) = 0. \tag{13}$$

Division by two appears only in this proof, not in the final circuit. Low and high blocks are disjoint, so the conjunction of (12) and (13) is exactly $\tau_2(S_c, T_c) = 0$.

Lemma 4.2 (Central-binomial carry identity)**.** *If $N = 2^e$ and $0 \le S, T < N$, then*

$$N^2 \mid \binom{2X}{X} \iff \tau_2(S,T) = 0, \qquad X = (N-1)((S+T+1)N + T + 1).$$

Proof. The exact block expansion is

$$X = (S+T)N^2 + (N-1-S)N + (N-1-T).$$

The lower blocks are e-bit complements. Thus $\mathrm{pc}(X) = 2e - \tau_2(S,T)$, while Legendre's factorial formula gives $\nu_2\binom{2X}{X} = \mathrm{pc}(X)$. Since $\tau_2 \ge 0$, divisibility by 2^{2e} is equivalent to its vanishing. □

This proves the reverse source implication. For the forward implication, start with a positive parent solution $\mathbf{p}$, choose an arbitrarily large $b = 2^{2q} > \max p_i$, and set $f = (b-1)/3$ and $g = \sum (p_i - 1)\mathcal{B}^{w_i}$. The all-one parent assignment is not a zero: its fifth residual is 1. Thus $g \ge \mathcal{B} > b$. Also

$$g \le (b-1)\sum_{j=1}^{t} \mathcal{B}^j < b\mathcal{B}^t = \mathcal{C},$$

because $b < \mathcal{B}$. Both carry tests pass. This proves Proposition 4.1 in full.

5 The single equation in ten unrestricted integers

Use exactly the ten integer unknowns

$$f, g, h, k, l, w, x, y, z, m.$$

Starting with (11), define the following polynomial abbreviations. They are the inherited Sun/package-31 finalizer [10, 3, 5] with the newly proved source substituted:

$$\begin{aligned}
L &= lY, & U &= 2LX, & V &= 4gwY, & W &= bw,\\
A &= U(V+1), & B_1 &= 2X+1, & C &= B_1 + (A-2)h,\\
P_n &= U^2V, & K &= X + 1 + k(P_n - 2),\\
S &= 2A - 5, & T &= 3WC - 2(W^2-1),\\
R_2 &= (P_n^2 - 4)K^2 + 4.
\end{aligned} \tag{14}$$

Next put

$$\begin{aligned}
\Delta_b &= 8\mathcal{C}^3 gK^2 - g^2(32(C-KL)^2\mathcal{C}^3 + g^2K^2),\\
O &= f^2l^2\Delta_b, \qquad \xi = zx,\\
D &= (A^2-4)C^2 + 4, \qquad E = C^2D\xi,\\
F &= 4(A^2-4)E^2 + 1,\\
G^* &= 2(A+2)(A-2)^2E^2 - 1, \qquad \Omega_0 = 16E,\\
H^* &= C + (B_1 - C)F + \Omega_0 Fy,\\
I^* &= ((G^*)^2 - 1)(H^*)^2 + 1.
\end{aligned} \tag{15}$$

The divisibility merger and radicands are

$$\Sigma = \xi S^5, \qquad r = T + Sy,$$

$$\begin{aligned}
&\Theta = (r^3 - rS^2 - S^3)(r^2 + 23S^2),\\
&r_0 = b(1 + (3O - 4)b^2z^2), \qquad r_1 = R_2, \qquad r_2 = DFI^*,\\
&v_1 = r_0, \qquad v_2 = r_0(P_n^2 + K^2 + 3).
\end{aligned} \tag{16}$$

Finally define

$$\begin{aligned}
&Z = \Sigma m - \Theta,\\
&a_0 = \Sigma^2 r_0, \qquad a_1 = \Sigma^2 v_1^2 r_1, \qquad a_2 = \Sigma^2 v_2^2 r_2,\\
&V_n = Z^2 + a_2 - a_0 - a_1, \qquad W_n = V_n^2 + 4Z^2 a_2 - 4a_0 a_1.
\end{aligned} \tag{17}$$

The promised equation over *all integers* is

$$\boxed{P_{10} = W_n^2 - 16V_n^2 Z^2 a_2 = 0.} \tag{18}$$

There are no roots, variable exponents, order predicates, or hidden parameters in this polynomial. The coefficient kernel (6), the explicit weights and targets, and the four fixed payloads determine every coefficient. The literal circuit is `data/polynomial.json`.

6 Proof that the finalizer preserves solvability

The numerical Lucas and binomial bridges are inherited from Sun's 2021 paper and 2026 preprint [9, 10]. Their use here is through their proofs with the source interface replaced by Proposition 4.1, as in package 40. All required source properties are now established: arbitrarily large square dyadic bases, the two binomial implications, the broad range bounds, $b = 1 + 3f$, the stated congruences and sizes, and $\mathcal{C} = 2^c b^d$ with fixed even c and odd d. No old quartic digit identity is used after that substitution.

6.1 The exact bridge obligations

Let $u_n(A)$ be the Lucas sequence $u_0 = 0, u_1 = 1$, $u_{n+1} = Au_n - u_{n-1}$, extended to integer indices. The cited bridge gives these implications for (14)–(15):

1. A positive parent solution yields witnesses with $A > 2$ even, $B_1 > 0$ odd, b square, $b \le g < \mathcal{C}$, $C = u_{B_1}(A)$, R_2 square, $S \mid T$, and $O > 0$.
2. Conversely, b square, $O > 0$, $\xi \neq 0$, DFI^* square, R_2 square, and $S \mid T$ imply a positive parent solution.
3. If $A > 2$ is even, $B_1 > 0$ is odd, and $C = u_{B_1}(A)$, then for every positive modulus divisor d one can choose $\xi > 0$ divisible by d so that every residue class modulo ξ contains a y for which DFI^* is a square.

The last statement is Sun's full residue-class interpolation theorem. In the reverse direction, $O > 0$ first gives $f, l \neq 0$, $g > 0$, $K \neq 0$, $g < 2\mathcal{C}$, and $4(C - KL)^2 < K^2$. The broad source bounds then make the Lucas reverse theorem applicable. It recovers $W = 2^{B_1}$ and the binomial divisibility; since $W = bw$ with $b > 0$, it also recovers that b is a power of two. This avoids assuming the digit interpretation before it has been justified.

6.2 Divisibility and nonvanishing in one expression

The quintic

$$\pi(v) = (v^3 - v - 1)(v^2 + 23)$$

has a root modulo every positive integer but no rational root. For an odd prime other than 23, either the cubic has a simple root, or it is irreducible and its discriminant -23 is a square, so the quadratic has a simple root. At 23 the cubic has the simple root 3. At 2, $-23 \equiv 1 \pmod 8$ is a square in $\mathbb{Z}_2$. Hensel lifting and the Chinese remainder theorem give roots modulo arbitrary positive integers. The rational-root test and positivity of $v^2 + 23$ exclude rational zeros.

The odd integer $S = 2A - 5$ is nonzero. If $S \mid T$, writing $T = Sq$ gives $\Theta = S^5\pi(q + y)$. A root of π modulo $|\xi|$ therefore permits a residue class of y with $\Sigma \mid \Theta$. Conversely, if $S^5 \mid \Theta$ but $S \nmid T$, choose a prime with $\nu_p(S) = \alpha > \beta = \nu_p(T)$. The two homogeneous factors of Θ have valuations 3β and 2β, contradicting $5\alpha \le \nu_p(\Theta)$. Also $\Theta \neq 0$, since otherwise $(T + Sy)/S$ would be a rational root of π.

6.3 The square and positivity conditions

For $z \neq 0$, if r_0 is a square then b and $1 + (3O - 4)b^2z^2$ are coprime. The second factor cannot be zero: that would force $|b| = |z| = 1$, hence $b = 1$, $f = O = 0$, a contradiction. Unique factorization makes b a signed square. A negative nonzero square is not 1 modulo 3, so b is a positive square. If $O \le 0$, the other factor is at most $1 - 4(bz)^2 < 0$, impossible. Thus $O > 0$.

Conversely, positive bridge witnesses have $O \ge g \ge b \ge 4$. The positive nonsquare $3O - 4 \equiv 2 \pmod 3$ has a Pell unit. Raising a positive unit to a suitable finite order modulo b produces $u^2 - (3O - 4)v^2 = 1$ with $v > 0$ divisible by b. Taking $z = v/b$ makes r_0 a square. Interpolation subsequently chooses ξ divisible by z, so $x = \xi/z$ is an integer. Its residue-class clause allows the same y to satisfy the quintic merger and the Lucas square condition.

6.4 A separating norm detects all three squares

For every assignment with $z \neq 0$, one has $|r_0| > 1$. Indeed $r_0 \in \{0, 1, -1\}$ would force $|b| = 1$, hence $b = 1, f = O = 0$, but then $r_0 = 1 - 4z^2 \le -3$. Also

$$\sqrt{|r_1|} \le P_n^2 + K^2 + 2.$$

Squaring this bound and subtracting $(P_n^2 + 4)K^2 + 4$ leaves $P_n^4 + K^4 + P_n^2K^2 + 4P_n^2 \ge 0$. Consequently

$$|v_1| > \sqrt{|r_0|},$$
$$|v_2| > \sqrt{|r_0|} + |v_1|\sqrt{|r_1|}.$$

If a signed weighted sum of the roots of r_0, r_1, r_2 is rational, all three roots must be rational. Otherwise take a Galois automorphism changing a root and consider the largest-index changed nonzero term. Its change dominates the sum of all earlier changes by these inequalities. The argument uses complex absolute values and also covers negative radicands. Rational square roots of integers are integer roots, so every radicand is a nonnegative square.

The exact identity

$$W_n^2 - 16V_n^2Z^2a_2 = \prod_{\varepsilon \in \{-1,1\}^3} (Z + \varepsilon_0\sqrt{a_0} + \varepsilon_1\sqrt{a_1} + \varepsilon_2\sqrt{a_2}) \tag{19}$$

follows by repeated multiplication of conjugates; the radicals cancel. An independent sparse expansion checks all 35 terms of the resulting degree-eight polynomial.

Given a positive parent solution, the preceding construction makes all r_i squares and ensures $\Sigma \neq 0$, $\Sigma \mid \Theta$. Choose m so that one factor of (19) vanishes. Conversely, if (18) holds and $\Sigma = 0$, its left side is

$\Theta^8 \neq 0$, a contradiction. Thus $\xi = zx \neq 0$. A vanishing norm factor divided by Σ is a rational weighted root sum. Separation makes all r_i squares, and integrality gives $\Sigma \mid \Theta$. The merger gives $S \mid T$, and the coprime argument gives b square and $O > 0$. The reverse bridge and Proposition 4.1 now recover the parent solution. Composition with the inherited parent theorem proves (1).

7 Exact degree, circuit size, and what the audits prove

The main circuit has 422 multiplication gates, 138 addition gates, and 80 subtraction gates. Every counted gate is reachable from its output; all ten input variables occur. Fixed powers are expanded into binary multiplications. Reuse is free, coefficient leaves are fixed inputs, and every binary operation in the delivered evaluator is counted.

The main source degrees are

$$\deg X = 34050, \qquad \deg Y = 22700, \qquad \deg \mathcal{C} = 351.$$

The inherited finalizer degree computation gives

$$456(34050) + 896(22700) + 24(351) + 1760 = 35\,876\,184. \tag{20}$$

This is independently certified on the actual DAG: degree propagation provides an upper bound, and after all ten variables are replaced by one indeterminate T, the coefficient at that bound is

$$[T^{35\,876\,184}]P_{10}(T, \ldots, T) \equiv 996362474 \pmod{1\,000\,000\,007}.$$

The nonzero residue proves a matching lower bound. This is exact coefficient arithmetic, not a random evaluation used to infer degree.

Claim	Evidence
Raw parent records	Symbolic reconstruction from all 22 original residuals, with the four ports left indeterminate.
Selective extraction	Exact symbolic equality of all 22 target coefficients, including all cross-row exclusions.
Fixed coefficients	Payload hashes, bit lengths, coherent tuple binding, exact coefficient bound, and complete reconstruction from the literal K215 transitions.
Final norm identity	Independent 35-term sparse polynomial expansion.
Operation count and exact degree	Full DAG liveness and degree/leading coefficient certificates.
Literal source/finalizer implementation	Independent modular replay of the stated formulas at signed sample inputs; diagnostic support in addition to the mathematical derivation.
Unbounded solvability equivalence	The new source proofs, the stated arithmetic bridges, and the inherited positive-parent theorem.

The auditors also reject deliberately changed weights and targets. The finite audits pass both normally and with Python optimization enabled. They do not search for a solution, prove consistency, or replace an unbounded compiler theorem. The distinction between exact symbolic checks and diagnostic sample replay is explicit in the receipts.

8 Nine positive unknowns: a second improved equation

The same support-selective source also improves the nine-variable Jones route. Its primary circuit has nine strictly positive unknowns, exact total degree 19 615 180, and 606 binary arithmetic operations. The equivalent version over nonnegative integers uses 609 operations. The previous package-45 positive equation has degree 129 875 980 and 788 operations. This is a reduction of approximately 84.9% in degree and 23.1% in operations, without changing the number of unknowns.

Let the raw quadratic source, its weights and its targets be as above, and write $W = 175$, $H = 5148$, and $K = H + 2W = 5498$. Set $M = 2^{2140017}$ and $b_0 = 2^{2140030} = 8192M$. Initially take $\epsilon, \phi, n \geq 0$ and $g, h, s, w, i, j > 0$. Define

$$b = b_0 + \epsilon, \qquad B = b^3, \qquad \ell = \sum_{r=1}^{31} B^{w_r},$$
$$N_0 = B^{W+1}, \qquad N_1 = 4B^{K+1}, \qquad N = N_0 N_1,$$
$$L = N_0 - 1 - (b-1)\ell, \qquad Z = \sum_{r=1}^{22} B^{E_r},$$
$$T = L + (B-2)ZN_0,$$
$$J_0 = 2(1+\ell+g)^2 \mathcal{D}(B) + B\sum_{r=0}^{K} B^r,$$
$$S = g + J_0 N_0, \qquad R = (N-1)((S+T+1)N + T + 1),$$
$$G_{\mathrm{cap}} = 32bB^W.$$

Every exponent is fixed. The raw residuals are evaluated at positive digits $x_r = z_r + 1$: the one added ℓ implements all thirty-one positivity shifts in the common convolution factor.

The coefficient at the largest exponent of $\mathcal{D}$ is 2, and all its other coefficients have absolute value below M. Hence $\mathcal{D}(B) > 0$ unconditionally. The source therefore satisfies $R, N \geq 8$ and $b \leq R, N$ before any equation is imposed. If $g < 4bB^W$, then $1 + \ell + g < 5bB^W$, and the choice $b \geq 8192M$ gives $0 < S, T < N$. With b a power of two, the exact population-count identity

$$\mathrm{popcount}(R) = 2\log_2 N + \mathrm{popcount}(S+T) - \mathrm{popcount}(S) - \mathrm{popcount}(T)$$

shows that $N^2 \mid \binom{2R}{R}$ is precisely the no-carry condition. Its low block encodes $g = \sum z_r B^{w_r}$ with $0 \leq z_r < b$; its target blocks impose all the original positive quadratic equations simultaneously. Conversely, every positive parent solution permits arbitrarily large dyadic bases with $32g < G_{\mathrm{cap}}$. The all-one parent assignment is not a zero, because its residual $a - E - sN$ equals -1, so $g > 0$.

All the numerical Jones bridge registers are as follows [6, 11]. The names K_1, D_1, H_1 distinguish these registers from the source exponent bound K, kernel $\mathcal{D}$, and maximum target H:

$$U = N^2 w, \qquad Y_0 = N^2 s, \qquad M_0 = RY_0,$$
$$P = 2M_0^2 U, \qquad K_1 = R + 1 + h(P-1), \qquad A = M_0(U+1),$$
$$B_1 = 2R + 1, \qquad W_1 = bw, \qquad C = B_1 + W_1 + \phi,$$
$$J = 4A - 5, \qquad Q = 3W_1 C - 2(W_1^2 - 1),$$
$$D_1 = (A^2-1)C^2 + 1, \qquad E = iJD_1C^2,$$
$$F = (A^2-1)E^2 + 1, \qquad G = A + F(CD_1 + 1 - A),$$
$$H_1 = B_1 + jC, \qquad I = (G^2-1)H_1^2 + 1,$$

$$a_0 = (P^2-1)K_1^2+1, \qquad a_1 = D_1FI,$$
$$u = JF, \qquad v = QF + J(H_1 - C).$$

Define the margin polynomial

$$m = K_1^2 G_{\mathrm{cap}} - 4(C - K_1Y_0)^2 G_{\mathrm{cap}} - 8gK_1^2.$$

Here m is a polynomial register, not a tenth unknown.

The hypotheses of `Jones1982.Nine.ratio_sound` and `ratio_complete` are $R, N \geq 8$, $b \leq R, N$, $b, G_{\mathrm{cap}} > 0$, $h, s, w, i, j > 0$ and $\phi \geq 0$. All have been supplied independently of the carry condition. Soundness states that a_0, a_1 squares, $u \mid v$, and $m > 0$ imply that b is a power of two, $N^2 \mid \binom{2R}{R}$, and $8g < G_{\mathrm{cap}}$. The latter is exactly $g < 4bB^W$, the interval needed by the source. Completeness starts with b a power of two, the divisibility, and $32g < G_{\mathrm{cap}}$, and supplies h, s, w, i, j, ϕ satisfying the four arithmetic conditions. This is precisely the stronger interval supplied by a parent zero. No property of the Sun ten-integer bridge is used here.

Set

$$V_0 = PK_1, \qquad V_1 = (AC)(AE)(GH_1), \qquad M_d = Q + H_1.$$

These registers are positive. The identities

$$V_0^2 - a_0 = K_1^2 - 1 > 0, \qquad (AC)^2 - D_1 = C^2 - 1 > 0,$$
$$(AE)^2 - F = E^2 - 1 > 0, \qquad (GH_1)^2 - I = H_1^2 - 1 > 0$$

give $0 < a_0 < V_0^2$ and $0 < a_1 < V_1^2$ on every outer assignment. Also $Q > 0$, $H_1 - C > 0$, and $0 < v/u = Q/J + (H_1 - C)/F < M_d - 1$.

One polynomial combines these four conditions. Put

$$\Omega = um, \qquad \Theta = u(n + M_d) + v - \Omega(2M_d + V_0V_1),$$

and take

$$\boxed{\left(\Theta^2 - \Omega^2(a_0 + V_0^2 a_1)\right)^2 - 4\Omega^4 V_0^2 a_0 a_1 = 0.}$$

At $m = 0$ its left side is nonzero. Otherwise its four-factor norm identity implies that a rational number is $r_0 + V_0 r_1$, with $r_t^2 = a_t$. The strict majorant excludes a zero rational sum; the identity $2(r_0 + V_0 r_1)r_0 = (r_0 + V_0 r_1)^2 + a_0 - V_0^2 a_1$ then makes both roots integers. Their bounds give $2M_d + V_0V_1 + r_0 + V_0 r_1 \geq 2M_d + 1 > 0$, which forces $m > 0$ and $u \mid v$. The converse chooses n from this same positive expression. Thus this finalizer has both the required soundness and completeness.

To define the polynomial on nine *positive* coordinates, replace ϵ, ϕ, n throughout by their positive coordinates minus one, leaving g, h, s, w, i, j positive. For the nine *nonnegative* version, instead leave ϵ, ϕ, n unchanged and replace each of g, h, s, w, i, j by its nonnegative coordinate plus one. These are bijections with the mixed domain used in the proof. They establish the stated solvability equivalences; they do not justify an unrestricted-integer interpretation of this nine-variable equation.

The full exact degree is

$$1152t + 12W + 280 = 19\,615\,180, \qquad t = 3(H + 3W + 2) = 17\,025.$$

The cancellation-resolved circuit proves this upper bound, and its all-variable diagonal has nonzero leading coefficient 2^{2050}, proving equality. Its critical exact identity is

$$\Delta_V = V_1^2 - a_1 = (C^2 - 1)(AE)^2(GH_1)^2$$

$$+ D_1(E^2-1)(GH_1)^2 + D_1F(H_1^2-1).$$

Writing $E_0 = u(n+M_d)+v-2\Omega M_d$, the inner polynomial of the final equation equals

$$E_0^2 - 2E_0\Omega V_0V_1 + \Omega^2V_0^2\Delta_V - \Omega^2a_0.$$

Its uniquely highest degree is $576t+6W+140$, with diagonal coefficient 2^{1025}. The final subtracted term has smaller degree than its square. The full proof is in `nine/PROOF.md`; the polynomial identity is independently replayed by the verifier. The source is explicit, while the finite audits are implementation checks, not a decision of consistency or a new end-to-end formal proof of the inherited compiler.

9 Reproducibility, formalization, and remaining opportunities

The archive is self-contained for all newly introduced finite algebra and circuit checks. It includes this LaTeX source and compiled PDF, literal circuits for both domains, fixed coefficient bytes and recipes, the raw parent, inherited proof excerpts and notices, source hashes, and independent audit programs. The main commands from the extracted package root are

```
python3 verify.py
python3 code/build.py
python3 code/audit_candidate.py
python3 nine/build_nine.py
python3 nine/verify_nine.py
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The first command performs the finite release checks in a temporary copy. The manuscript can be incorporated as a new report directory while preserving the supplied licenses and pinned provenance.

The most useful next formalization step is the generic selective extraction lemma with a finite certificate as input. Its proof is a short multinomial coefficient calculation plus finite noninterference conditions. A second small module can prove the single-word positivity substitution. Those lemmas can be linked to the existing data-valued compiler infrastructure without using a proposition-valued existence theorem to fabricate executable coefficient data.

The main unresolved end-to-end proof-assistant task remains the inherited native-window compiler bridge. A checked arithmetic equivalence about syntactic ZFC consistency is also distinct from a derivation in the object-language ZFC proof calculus. This report supplies ordinary mathematical reductions intended for that formalization; it does not relabel either open task as completed.

There remain concrete optimization problems: jointly choosing selective weights and targets, exploiting equality of coefficient vectors rather than requiring every supported exponent to be unique, and choosing a different sparse quadratic parent. The certificate formulation makes such experiments inexpensive to validate. Any future reduction must continue to check all unused monomial contamination and all broad witness bounds. Those two requirements are essential to the present proof.

A A modest additional local degree-68 improvement

The active repository's checked 75-operation degree-68 polynomial uses fifteen positive variables and the six factors $N_0, N_1, N_a, N_k, N_t, N_s$ of its `SHORT_SYSTEM.md`. Its equation is

$$(N_0-1)^2 + (N_a-1)^2 + (N_s-1)^2 + (N_1N_kN_t-1)^2 = 0.$$

A new local alternative is

$$N_0^2 + N_a^2 + N_s^2 + (N_1 N_k N_t - 1)^2 - 3 = 0. \tag{21}$$

It has the same positive zero set and costs 74 operations at the same exact degree 68. This is a one-gate local improvement, separate from the main low-variable construction.

The reason is that N_0, N_a, N_s are nonzero before any equation is assumed. The first and third are norms with nonsquare discriminants $n(n+1)$ and $(a+2)^2 - 1$, and their relevant positive coordinates are nonzero. For the middle factor write $N_a = AV^2 - (A-1)y^2$ with $A = \Delta(f^2-1)$. If $f = 1$, it equals $y^2 > 0$. Otherwise $A \geq 2$; a zero would make the reduced fraction $(A-1)/A$ a rational square, forcing both consecutive positive integers $A-1, A$ to be squares, impossible.

Thus the first three squares in (21) are at least one; equality forces each to be one and forces $N_1 N_k N_t = 1$. The inherited negative Pell and modulo-four unit-sign arguments force $N_0 = N_a = N_s = 1$. Conversely those values make (21) vanish. Only N_s^2 reaches degree 68, with leading form

$$i^4 w^4 s^{12} (\eta + \zeta)^8 (\beta J)^{40}.$$

The local JSON, proof, and finite audit are supplied under `short/`. This is an ordinary local proof; it is not represented as a newly compiled Lean theorem. Its source tuple and coordinate conventions are preserved separately from the main package-31 construction.

B A smaller complete native coefficient tuple

A separate adapter in `coefficient_adapter/` reduces the fixed coefficient heights of the inherited K215 native/Pell interface:

$$\rho = 2^{89}, \qquad L = 4831, \qquad B = \rho^L = 2^{429959}.$$

Its four raw coefficient ports occupy 215 492 bytes, versus source44's 295 388 bytes for $B = 2^{590069}$, a reduction of 27.0478%. The base exponent decreases by 27.1341%. The port bit lengths, in order $(\beta, K, M_C, M_{F,\mathrm{source}})$, are $429959, 434054, 429959, 429960$. The main low-variable equations in this article retain their source31 tuple. Substituting the new tuple would require rebuilding their derived index constants.

The new mechanism. Source40 factors 488 receiver tiles into two modes and seventeen Boolean payload bits, leaving 457 selector modes, but explicitly leaves an empty-cell soundness obligation open. Source44 supplies aggregate complement clauses. The new adapter combines these ideas with complemented edge labels, doubled horizontal labels, and reuse of the copied west parity bit as the current anchor. It removes four dedicated occupancy copies and the separate current anchor; the support has 520 native positions.

Put $T = 2^{42}$, $M = 4T$. The selector and gate masks give $n \in \{0,1\}$. For mode $i = (w_i, n_i, e_i, s_i)$ and receiver payloads $Q_R, Q_L \in \{0, \ldots, 255\}$, $u \in \{0,1\}$, define

$$\begin{aligned} L_i &= 2w_i + 2^{11} n_i + 2^{21}(2e_i) + 2^{32} s_i, \\ V &= 2Q_R + 2^{11} u + 2^{21}(2Q_L) + 2^{32}(2Q_R + 2Q_L + u). \end{aligned}$$

The label widths are $11, 10, 11, 10$. Let $Z < T$ be the copied label vector, $y = Z \bmod 2$ its west low bit, and A_1, A_2 the two standalone anchors. The same physical bit y is used twice:

$$W = Z + T(y + A_1 + A_2), \qquad \sum_i x_i(1 + L_i) + V + W \equiv 0 \pmod{M}. \tag{22}$$

All coefficients are nonnegative. Every L_i, V is even, so (22) gives $y = n$.

If $n = 0$, then $W < 3T$ and $V < T$. Hence $0 \le V + W < M$, so $V = W = 0$: all payloads, copies and anchors vanish, apart from the ignored dummy digits. This closes the old empty-cell gap. If $n = 1$, the gates give $0 \le L_i + V < T$, and the positive sum in (22) is less than $2M$. It therefore equals M, giving

$$W = 3T + \big(T - 1 - (L_i + V)\big).$$

Thus $y = A_1 = A_2 = 1$, and the copied labels are exactly complemented. Both horizontal labels were doubled, so their complemented low bits are one. Conversely every intended occupied encoding satisfies the clause. This uniform interval argument supplies the local theorem.

The full native interface. The shared source anchor is the west copy at position 539; the two tested temporal anchors remain $1203, 1807$. The bare shifts are $33, 68, 664, 1268$, clause targets $2409, 3015$, and spatial exponent 46. Complete basis identities and the two exact support intersections give a clean Start clause, binary synchronization including the upper-dummy exception, and whole-cell alignment. Spatial matching then propagates occupation through the cyclic word: an occupied east low bit is one, while the west low bit equals n. Complementation and doubling preserve edge matching. Deleting the 536 unmatched receiver expansions recovers the original marked K215 Wang language.

The masks have populations 520 and $429959 - 520$, with the same required low congruences as the source interface. All conservative carry and strict-slack thresholds are below 2^{89}. Completeness uses the fixed prime $\lambda = 106261$:

$$\operatorname{ord}_{429959}(B) = 26565 \mid (\lambda - 1), \qquad B^{\lambda-1} \bmod \lambda^2 = 8376873414 \neq 1.$$

The certified controller units allow the inherited Boolean dummy adjustment of the actual packed rank. The full coefficient recipe, synchronization proof, completeness argument and canonical witness promises are in `SEPARATED_PROOF.md`.

Verification and scope. The independent auditor imports neither builder and reconstructs the Wang language from all 430 literal K215 actions. It verifies 37 960 native basis identities, 1 479 occupied encodings, 65 076 physical bit corruptions, and all 131 072 empty auxiliary assignments, of which only the zero payload is realizable. Every coefficient byte and the controller arithmetic are independently reconstructed. Normal and optimized Python executions pass.

This is a complete ordinary mathematical adapter composed with the explicitly inherited native/Pell compiler, Wang-language theorem, and K215/refutation calibration. It is not a new end-to-end proof-assistant certificate. Finite checks establish the stated finite identities; the universal transfer uses the proof and its explicit inherited theorems. Exact source pins and original licenses accompany the files.

C Attribution and source boundary

The new contributions in this report are the instantiated selective separation condition, the explicit span-175 certificate, the shared positive-word substitution, nonuniform target arrangement and common translation, their source proof, their compositions with the inherited finalizers, and the resulting literal circuit and exact-degree audits. The local P74 argument is an additional algebraic observation. Appendix B gives a separate reduction of fixed coefficient heights using a parity anchor and a complete empty-cell decoder. It does not alter the tuple used by the main two equations.

The K215 machine, native tile compiler, coefficient tuple, quadratic parent, Sun and Jones bridges, and final norm constructions are inherited and attributed. Package 40 had already identified support-sensitive separation and nonuniform targets as research directions; this report realizes that direction with an explicit certificate and full source argument. It does not claim global priority for sparse coefficient coding or for the general nine-/ten-variable theorems. Files under `upstream/` retain their original mathematical claims and dates as provenance, not as new achievements of this report.

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